SAQ - Paper-I

(Group-A)

GENERAL PHYSIOLOGY

Introduction

1. Define homeostasis. **(CMU**: Nov-22, May-20) (Ja-12, 14, 15, 16, Nov-17,18)
2. Mention its composents. Explain with example how negative feedback maintain homeostasis. **(CMU: May-25)**
3. Show in a table the different constituents of ECF and ICF with their normal values and non-lethal limits. (Ju-09, Ja-14)
4. Define Gain **(CMU:** Nov-24, 18, May-23) **(CMU: Nov-25)**
5. Name systems regulated homeostasis.
6. Compare & contrast between negative feedback and positive feedback mechanism. **(CMU:** May-23, **Nov-24,** 22) (Ju-07,Ja-15, Nov-18)
7. Negative feedback mechanism (May-15)
8. Discuss diagrammatically that positive feedback is vicious cycle (Nov-15)
9. **S/N:**
   a) Feedback system (July-15) b) Homeostasis **(CMU:**May-19) (Jan 16, Nov-16)

Feedback System

1. Positive feedback mechanism can sometimes enter into a vicious cycle and causes death. explain? **(CMU:**May-20) (Ju-09, 11) Discuss diagrammatically that positive feedback is a viscious cycle. **(CMU: Nov-25)**
2. Explain- Most of the body functions are regulated by negative feedback mechanism? (Ja-08 Nov-17)
3. Give the flow chart of negative feedback mechanism for controlling homeostasis **(CMU:** Nov 19) (Ja-14)

Cell & Cell Membrane

1. Show in a diagram components of cell membrane and mention the function of its components (Ju-05, Jan-16,17 May-16)
2. List the intercellular connection. Describe any two of them. (Ju-08, Ja-09, 12, May-15) Illustrate the functional importance of this Junction? (May-15)
3. Draw and level a typical cell membrane. How it maintains the cell volume?
4. Mention the functions of mitochondria (Ja-13). Describe in brief about Mitochondria (May-18)
5. Name the different organelles of a cell.
6. The lysosome provides the intracellular digestive system. **(CMU:Nov-23)**
7. Write down the function of cell membrane protein (May-18)
8. List the membrannous organelles **(CMU:** May-21, Nov-19)
9. **Short note:** a) ER, Lysosome (Ju-12) **(CMU: Nov-25)** b) Golgi apparatus. **(CMU:**Nov-19 c) Mitochondria (Nov-18, Ju-13)
10. Discuss the function of mitochondria and golgi apparatur with diagram. **(CMU: May-25)**

Membrane Transport

1. Write down difference between secondary active transport and facilitated diffusion? **(CMU**:Nov 19) (Ja-11)
2. Name the energy dependent transport processes across the cell membrane with example **(CMU** Nov-21, 20)

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3. What do you mean by carrier mediated transport? (May-16) Importance of carrier mediated transport process. (Ju-10, 13)
4. Name the carrier mediated transport process. **(CMU:** May-24) Give the role of Na^+-K^+ ATPase in physiology. **(CMU:** May-22) (Ju-10) What do you mean by cell sickness syndrome? (Ja-07)
5. What do you mean by Na^+-K^+ pump? Give the importance of it. **(CMU:** Nov-20) How can this pump be blocked? (Ja-14) Explain the rule of Na^+- K^+ pump in physiology **(CMU:** May-24) Explain how cell volume is maintained by Na^+ K^+ pump. **(CMU: MNov-25)**
6. Name the different ion channels on cell membrane. Explain the clinical importance of these channels.
7. What is membrane transport, classify it. (Ja-13) with example. **(CMU: Nov-25)**
8. Define symport and counter transport? Difference between Osmosis and diffusion. (Ja-13, Nov-16)
9. Differences between active transport and passive transport. **(CMU:**May-20) (Ja-05)
10. Write the process of endocytosis (May-16) & exocytosis (Ju-13)
11. **SN:** Na^+-K^+ pump. (Ja-15, Nov.15)
12. Discuss briefly primary & secondary active transport with example **(CMU:**May-22) importance. (Ju-14, May-15)
13. Difference between osmosis and active transport (July-15)
14. S/N: Active transport **(CMU:**May-19) (Nov-18, May-19)
15. Define and classify biotransport system **(CMU:**Nov-23, 20, 19)

Membrane Potential

1. Name the different types of membrane potential. Briefly describe the different phases of action potential **(CMU**:Nov-19) What is end plate potential? **(CMU:**May-23)
2. Briefly discuss the ionic basis during an action potential. **(CMU:**May-20) (Ju-11)
3. Define action potential. **(CMU:** Nov-24, 23,18) How it differs from pace maker potential and graded potential. (Ja-06, Ju-11)
4. Draw and lable a typical action potential **(CMU:**May-20) (Ja-14) How is it propagated through a nerve fiber? **(CMU:**Nov-23, 22) (Ju-09, 13,14, May-17) with figure **(CMU:** Nov-24)
5. Explain physiological basis/ Ionic basis of action potential? (Ja-07) (Nov-18) How can it be block (Nov-18)
6. Differences between action potential and graded potential. (Ja-07, Ju-11)
7. Define RMP. How is it maintained **(CMU:** May-23, 15, July-15) & re-established? (Ja-05, 07, 14). generated **(CMU:** May-24). Why RMP is negative? Resting membrane potential in different cell.
8. Show in a diagram the action potential in a cardiac muscle? What are the electrolytes responsible for action potential in cardiac muscle? (May-15, July-15)
9. Draw and lable an action potential in skeletal muscle (May-19) and cardiac muscle **(CMU:**Nov-20, May-19)
10. **S.N-** a) Action potential. (Ja-12) b) RMP (May-16)
11. Define Action Potential. Show in a diagram the channel responsible for Pacemaker Potential and Action Potential. **(CMU:**May-21)

Muscle

1. Elaborate the functional differences among the three kinds of muscles. (Ju-06)
2. State the properties of skeletal muscle. **(CMU:**Nov-19, May-22) (Nov-16) Explain-Why skeletal muscle could be fatigue but cardiac muscle never? (Ju-10) **(CMU:** May-24)
3. Mention the steps of skeletal muscle contraction with figure (Nov-16) & relaxation. **(CMU:** May-19) (Ju-14, Ja-17) or what is excitation contraction coupling of skeletal muscle and mention the steps involve in this process. **(CMU:** May-24) Mechanism of muscle contraction.(Ja-13) (Contraction relaxation) (May-17)

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4. What do you mean by isotonic & isometric contraction? Role of Ca^{2+} in muscle contraction.
5. What is myofibril? Differences between slow & fast fibre. What do you mean by motor unit?
6. Draw & level of a neuromuscular junction. (Ju-14)
7. Role of ATP in Muscle Contraction. (Nov-16)
8. Mention the sequence of events of smooth muscle contraction and relaxation. **(CMU:** Nov-20, (May-18)
9. **S.N-** a) Muscle twitch (Ja-07), b) Rigor mortis, c) Muscle tetany.
10. Explain how nerve impulse is conducted to muscle. **(CMU:** May-23, 22)

BLOOD

Introduction

1. Give the origin, normal value (Nov-18, Nov-15, Ja-16) & function of plasma protein **(CMU:** Nov-24, 23, 22, 20, 18, May-22)
2. Importance of plasma colloidal osmotic pressure. **(CMU:**May-22) (Ja-05, Ju-05, Ja-11, 12)
3. How can plasma protein be separated? (Ju-10, Ja-13, Nov-15, Ja-16, May-17)
4. State the importance of electrophoresis of plasma protein. (Ju-05, 06, Ja-12)
5. Name the plasma protein with their normal values. List the function of each specific plasma protein. **(CMU:** Nov-22, May-20) (Ju-07,08, Ja-15, 17) & How can plasma protein separated (May-17)
6. Discuss formed elements with their Normal blood values (Nov-16)
7. List the properties and function of Plasma Protein, what are the effect of hypoproteinemia **(CMU:** May-21) Effect (Nov-23)

R.B.C

1. Show in a diagram the different steps of erythropoiesis **(CMU:**Nov-22, May-19) (July-15, Nov-16, May-19). How the production of RBC is adjusted to meet the varying needs of the individual? (Ja-06) Define erythropoiesis and mention the changes occur during erythropoiesis. **(CMU:** Nov-24)
2. What are the basic changes that occur in erythropoiesis? Name the erythropoietic factors **(CMU:** Nov-22, 19, May-19) and mention their role in erythropoiesis. (Ja-09)
3. Describe the life cycle of R.B.C. **(CMU:**Nov-19) (Ja-10, May-15, July-15)
   Show the fate of RBC after 120 days in a flow chart. **(CMU: Nov-24)**
4. Define anaemia. State the morphological types of anaemia with example. **(CMU: May-25,** 19) (Ja-16)
   List the investigation you will suggest to assess a case of anaemia. (Ju-12) **(CMU: May-25,** 24)
5. Describe the steps of synthesis of Hb **(CMU:**Nov-20) (May-15). Why anemia occurs in vit -B_{12} and folic acid deficiency?
6. Discuss about factor of erythropoiesis ? (Ja-16)
7. Write down the function of RBC **(Nov-23,** May-18), Hb (May-15, July-15).
8. Function of neutrophils (Nov-17)
9. Briefly discuss about the metabolism of hemoglobin. **(CMU: May-24)**
10. Name some cause of pancytopenia (May-19)
11. Classify anemia with example. Mention the clinical investigation for diagnosis of anemia. **(CMU:** May-20)
12. Classify Hemoglobin with their functional differences. **(CMU: May-24)**
13. **S/N-** a) Erythropoiesis (Ja-14)
    b) Anaemia **(CMU:** Nov-22) (Ja-15, Nov-15)
    c) Polycythemia **(CMU:** Nov-19) (Ja-05)
    d) Erythropoietin (Ja-03)
    e) ESR (May-18)
    f) Machocytic Anemia. **(CMU:** Nov-21)

Megaloblastic anemia

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W.B.C

1. Give the differential and absolute count of W.B.C **(CMU:**Nov-20) (Cu-04)
2. What are the properties of WBC? Describe any two of them (May-16)
3. Write down the role of neutrophil in inflammation/acute response
4. Name the immune cell. Illustrate the functional importance of these cell in our body? (July-15)
5. What is Leucopoiesis ? (May-16) Show in a diagram different stage of Leucopoiesis (Nov-17)
6. Define chemotaxis. (May-18)
7. Steps of phagocytosis, what is the role of lysosome in phagocytosis. **(CMU:**Nov-21) (Nov-16)
8. Write difference between T and B Lymphocyte. **(CMU:**Nov-20)
9. SN- a) Macrophages (Ja-06) b) Thrombocytopenia (Ja-08)
   c) Pancytopenia (Ja-07) d) Leucopoiesis (Ju-09)
   e) Phagocytosis **(CMU:**Nov-19) f) Lymphocyte **(CMU:**May-20)
10. Write down the function of WBC. **(CMU: Nov-25)**

Platelets

1. Define hemostasis **(CMU:**Nov-22, 20) (May-18, Nov-18, Ja-15,16) State the role of platelet in hemostasis. **(CMU:** May-19) (Ju-06,11,13, Ja-05,13)
2. Define & Discuss the basic steps of hemostasis. **(CMU:May-25,** Nov-20) (May-19, Ju-14, Nov-15, Jan-16) Name the substances that release from platelet and mention their importance. (Ja-06, 10)
3. State the function of platelet **(CMU:**May-19) (Ju-12, 13, Ja-14,15, Nov-15, May-18)
4. What are the active factors present in the cytoplasm of platelets? **(**May-23**)**
5. Explain the role of platelets in vascular spasm. (Ja-08, Ju-08, May-19)
6. What is VON WILLEBRAND factor? What are the consequences of deficiency of this factor? (Ju-10)
7. Role of coagulation factor in common pathway for coagulation (Ju-12)
8. What is platelet activation? (Ju-14)
9. Name some bleeding disorder. (Ja-15)

Clotting System

1. Show the different steps of blood coagulation in intrinsic pathway**. (CMU:** Nov-22, 21) (Ju-12, Ja-17, May-16, 19) Mention the essential steps of clotting mechanism. **(CMU:** May-24)
2. Differences between the hemophilia and purpura. (Ju-11, 13)
3. Write basic steps of blood coagulation. **(CMU:**Nov-23, 19) (Ju-06). Why blood clotting become abnormal in patient with vit-K deficiency? (Ja-05)
   Name the vit-K dependant clotting factor. **(CMU:**Nov-23**)**
4. Define hemostasis. (Ja-06, 12, Ja-17, May-17) Describe mechanism of blood coagulation in a test tube. (Ja-05, 07, 08, 09,13, Nov-18)
5. Name the test should be done to diagnose a case of bleeding disorder. (Ju-09, Ja-14)
6. Discuss the extrinsic mechanism of blood coagulation. **(CMU:** May-21) (Ju-07, Nov-16)
7. Define coagulation? **(CMU:** Nov-21) (May-16)
8. How prevention of blood clotting is achieved in normal blood vessel? **(CMU:** Nov-20, 19) (May-15, Ja-16)
9. How platelet plug is formed? **(CMU:** May-21) (Ja-16, Nov-16) & importance of it **(CMU:**May-23**)**
10. Define fibrinolysis. State the mechanises of lysis of a clot. Why blood does not clot inside the blood vessel. **(CMU:** May-22) Explain how temporary hemostatic plug is formed **(CMU:**May-23**)**
11. Describe fibrinolytic mechanism after blood clotting.
    Explain how intravascular anticoagulant prevent blood clotting in normal vascular system. **(CMU:** May-24)

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12. Write about haemophillia (May-19)
13. Define Clot **(CMU:** May-21) Steps of clot retraction & its benefits **(CMU:**Nov-23)
14. **Short note:** a) Purpura (Nov-15, 17, Ju-15) b) Hemostasis (Nov-16)

Blood Group

1. Show in a table the percentage/frequency of different blood group in ABO system. (Ju-11, Nov-17)
2. Name the common blood types (Ja-07) & describe how blood is typed & cross matched? (Ju-06) What measure should be taken for safe blood transfusion? (Ja-06,Ju-12)
3. List the hazards of mismatched blood transfusion. **(CMU:** Nov-21, May-21, Nov-19) (Ja-07, Ju-13, May-16, 18)
4. State the Land Steiner's law. What do you mean by mismatched blood transfusion (Nov-17)? Why is there acute renal shut down in mismatched blood transfusion? (Ja-08)
5. Describe erythroblastosis fetalis. Enumerate the hazards of blood transfusion
6. Give the basic principle of blood grouping? **(CMU:** May-21) (Ju-13, May-18)
7. Indications for blood transfusion **(CMU:** Nov-21) (May-16)
8. How Rh incompatibility can be prevented? (Nov-17)
9. SN- a) Safe blood transfusion (Ja-08) b) Delayed reaction of mismatched blood transfusion. c) Blood transfusion (Ju-06, Ja-14) d) Rh factor **(CMU:**Nov-19) (Ja-09, 12,13) e) Exchange transfusion f) Blood grouping **(CMU:**May-19) (Ju-14, Ja-17) g) Mismatch blood transfusion (Ja-15, Nov-15, 16, 17, Ja-16) h) Erythroblastosis fetalis. **(CMU:**May-21, Nov-20) (Nov-18)
10. Name common blood types. Describe how blood is typed and cross matched. What measures should be taken for safe blood transfusion. **(CMU: May-25)**

Immunity

1. Short note- a. Autoimmunity (Ju-08) b. Leucopoiesis c) Immunity (Ja-17, May-17)
2. Difference between innate and acquired immunity. **(CMU: Nov-25)**

CARDIOVASCULAR SYSTEM (C.V.S)

Cardiac Muscle

1. Explain how impulse is generated in SA-node & transmitted to ventricular muscle with diagram. **(CMU:**May-23)
2. What are the properties of cardiac muscle? **(CMU:**Nov-24, 22,19) (May-18, 22) Describe any two of them. Discuss the effects of autonomic nerves on heart. **(CMU:** Nov-22) (Ju-09, Ja-11, 12, 14). Why cardiac muscle never fatigue? (Nov-19) Mention the physiological properties of ventricular muscles. **(CMU:** May-24)
3. Name the functional tissues of the heart. (Ju-07,13, Ja-15, Nov-15, 18, Ja-16) State the Frank Starling law with its importance. **(CMU:**Nov-19) (Ja-06,13, Nov-16) Name the pacemaker tissue. (May-17)
4. Show in a diagram the descending order of cardiac impulse transmission. (Ju-05, Ja-15, 17, Nov-15, 18) Discuss the effects of conduction block in different sites of this pathway
5. What is A-V nodal delay? What are the significances of A-V nodal delay? **(CMU:**May-23) (Nov-18, Ju-05,07,12, Ja-15, Nov-15, Ja-16)
6. Name the pace maker tissues.(Ja-17) Why S.A node is called pacemaker of the heart? (Ja-13) Discuss the relation between muscle length, tension and velocity of contraction of cardiac muscle. (Ju-05, 08, Ja-09)
7. How cardiac impulse is generated automatically? Write down the cholinergic and adrenergic stimulation on SA nodal membrane. (Ju-10)

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8. Draw & lable transmission of cardiac impulse from SA node (Ja-11,12)
9. Discuss the origin & transmission of cardiac impulse? (Ju-13)
10. Show in a diagram the descending order of cardiac impuls transmission. (Ja-16)
11. What are the roles of different Nerves on heart ? **(CMU:**May-21) (Nov-16)
12. What is refractory period **(CMU:**May-19)
13. State the laws followed during cardiac muscle contraction with their importance. **(CMU:** May-24)
14. **Short note:** a) Poiseuille's law (Ja-08) b) Pacemaker potential (Ju-09)
    c) Vagal escape (ja-06) d) Av nodal delay **(CMU:**Nov-22)

Cardiac Cycle

1. Define cardiac cycle. **(CMU:**Nov-21, May-20) (Ja-05, Ju-06,12, Ju-15) What are the changes that occur in cardiac cycle? (Ju-09, Ja-14,15, Ju-15)
2. Show in a diagram the various events of cardiac cycle **(CMU:** Nov-21) (Nov-18, May-17)
3. Write down the important features of period of isometric contraction. (Ju-07) Draw and lable the events of cardiac cycle at a heart rate of 75 beats/min (Ju-08) or 72 beats/min(Ju-12)
4. Outline the changes that occur in the heart during systole and diastole in a cardiac cycle. What do you mean by pre load and after load? (Ja-07,15, Nov-16)
5. Show in a diagram the pressure changes in the different chambers during cardiac cycle. (Ju-06, Ja-14, May-17)
6. Draw and level action potential of ventricular muscle and SA node of heart
7. Show in a diagram about various changes during ventricular systole & diastole. **(CMU:**May-20)
8. **SN:** Cardiac Cycle **(CMU:Nov-19)** (Nov-16, May-19)
9. A 30 years old male's heart rate is 90 beats/min. Calculate the cardiac cycle time. **(CMU:**Nov-21)
10. SN: Cardiac output. **(CMU: Nov-25)**

Heart Sound

1. Define & classify heart sound. Describe the significance of 1st and 2nd heart sound.
2. What are the abnormal heart sounds? What is Korotkoff sound.?
3. How heart sound are produced ? (Ju-15)
4. **S/N:** Heart sound **(CMU:**Nov-20)

Cardiac Output

1. Define cardiac output and cardiac index. Discuss the factors controlling it. (Ja-13, Nov-15, Ja-16) How cardiac output can be measured? **(CMU:**Nov-19) (Ja-06, 10, 11, 12,15)
2. How cardiac output is measured by applying Fick's principle? (Ja-10,11,15, Nov-15, May-18)
3. Factors influencing venous return. Relation between cardiac output & blood pressure(Ju-12)
4. What is Fick's Principle (Ja-16)
5. S/N: Ventricular escape (May-18)
6. **Short note-** a) Venous return. **(CMU:**May-22) (Nov-18) (Ju-07, May-19)
   b) ESV c) Cardiac output (May-19)

Heart Rate

1. Name the factors affecting heart rate. **(CMU:**Nov-22) (May-18)
2. **SN:** Heart rate. **(CMU: Nov-25) (CMU:**May-20)

E.C.G, Heart Block & Shock

1. Define and classify heart block. **(CMU:** May-19) (Cu-04, Ja-12, May-15, 19)
2. Define shock. Name its major types. (Ju-06, May-15)
3. **Short note-** Hypovolemic shock (Ja-07)

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4. Write few causes of hypovolumic shock (May-15)
5. Briefly discuss about normal ECG with it's interpretation. (May-18)
6. Write down the important feature of ECG. (Nov-18)
7. What information can we get from ECG tracing (May-19)
8. **S/N:** a) P-R interval. (Ju-08,09,13, Ja-13,14,15) b) ECG (Ju-14) c) Shock **(CMU:**Nov-19)
9. Draw & lable normal ECG. Mention the relationship of ECG to the Cardiac cycle. What are the information we get from PR & RR interval. **(CMU:** Nov-24, May-23, 22**)**
10. Define & classify shock with example. Briefly discuss about compensated stage of shock. **(CMU:** May-22**)**
11. From ECG how can we diagnose different types of AV-nodal black. **(CMU:**May-23**)**
12. Write the clinical importance of the intervals and ST segment. **(CMU:** Nov-24**)**

Blood Pressure

1. Define the blood pressure (Nov-23, Ju-13, May-19). Give the types or classification **(CMU:**Nov 23, 20**)** **(CMU:** May-20, 19**)** (Nov-15), normal range, **(CMU:** Nov-21), and significance of blood pressure. (Ja-06)
2. Define & classify pulse. **(CMU: Nov-23**, 22, May-21**)**
3. Define pulse & mean pressure with their normal values & importances (Ju-12)
4. What are the mechanism of regulation of blood pressure.(May-16, 19) Describe one of the **(CMU:**May-21**)**
5. Describe the short time regulation of blood pressure. **(CMU:** May-19**)** (Ju-14, Nov-17, 15, Ja-1
6. What do you mean by long term **(CMU:Nov-23)** regulation of blood pressure? Explain the mechanism of long term regulation of blood pressure. **(CMU:**Nov-20**)** (Ja-08, Nov-16)
7. Classify HTN (Nov-16)
8. State the role of buffer in blood pressure regulation. Show in a diagram integrated system of arterial pressure regulation. (Ja-07 Ju-06, 10, 11)
9. Discuss the role of kidney in arterial pressure or Blood Pressure regulation. **(CMU:** Nov-22, May-20**)** (Ju-05)
10. Outline the compensatory mechanism that maintain the blood pressure on rising from lying supine to the standing position. (Ju-13, Ja-06)
11. What do you mean by postural hypotension? (Ju-06) Reflexes regulation the blood pressure. (Ja-10)
12. Discuss in details the mechanism of regulation of arterial blood pressure when there is acute loss of 1 litre of blood. (Ju-08)
13. Explain the role of angiotensin-II in blood pressure regulation. Effect of exercise on blood pressure. (Ju-11)
14. What do you mean by peripheral resistance? Discuss the factors regulating peripheral resistance. (Nov-17
15. List the reflex mechanism for maintaing normal arterial pressure **(CMU:**May-22**)**
16. Explain the relationship among pressure, flow & resistance in the vascular system. **(CMU: May-24**
17. **Short note:** a) Baroreceptor (Ja-06, 11, Ju-09) b) Brain bridge reflex
    c) Renin -Angiotensin mechanism d) Pulse **CMU:** Nov-21)(Nov-15)

Microcirculation

1. Define the microcirculation. **(CMU:**May-19**)** (Ju-05, 11, May-15,17 Ju-15, Ja-17)
2. Write down the components of microcirculation. (May-16) State the mechanism of exchange of fluid and nutrition in the microcirculation bed. **(CMU:May-24,** 19**)** (Ju-12,15 Ja-17 May-15, 17
3. **SN:-** Sterlin's Law (May-16)

Vascular System

1. Classify blood vessel on physiological point of view **(CMU: May-24)** (Ja-01, May-18)
2. Name the hormones that cause vasodilatation and vasoconstriction, the arterioles. Write for variables that cause the net filtration pressure across the systemic capillary. (Ja-05, 08)

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3. Name the reflexes in cardiovascular regulatory system. Which of the reflexes is called pressure buffer system? Explain the pressure buffer action of this reflexes.(Ja-08)
4. Discuss the factors regulating tissue blood flow?(Ju-07)
5. What is the relationship between flow, pressure, resistance in vascular system. (Nov-17)
6. **Short note:-** a) Pulse (Ja-10) b) Coronary circulation (Ju-10)

PAPER-I

(GROUP-B)

RESPIRATORY SYSTEM (Respiration)

1. Define respiration. Name the muscles of respiration and their machanism of action **(CMU:** May-24, May-19) (May-19) (Ja-12,15, Nov-15, May-19)
   Mention the changes that occur during quite respiration **(CMU:**Nov-24, May-23)
2. Discuss the mechanism of breathing. (Ja-07, Ju-06, Nov-17) **(CMU:**May-19) Difference between alveolar and pulmonary ventilation. **(CMU:**Nov-20, Nov-19) (Ja-12, Ju-15, Nov-15, Ja-16)
3. List the respiratory & nonrespiratory functions of lungs. (Ju-03,13, Ja-15, Nov-15) **(CMU: Nov-25**, 22, May-19). How you calculate this **(CMU**: Nov-19)
4. Explain inspiration is an active process & expiration is a passive process. (Ja-09, 12)
5. Show in a diagram the intrapleural & intrapulmonary pressure changes in relation to atmospheric pressure during inspiration and expiration. (Ju-05, 06, Ja-07,11, May-15, 17, Ja-17)
6. Why intra pleural pressure is always negative? **(CMU:**May-20) Write down the effect of hyperventilation on vital organ. (Ja-05, 07, Nov-18)
7. Difference between minute respiratory volume and alveolar ventilation. **(CMU:**May-22) Briefly discuss the defensive function of lungs. **(CMU:**May-22) (Ja-10)
8. What is dead space air? Write down the difference between physiological & anatomical dead space air? (Ja-14)
9. What is periodic breathing. (Ja-16, Nov-16)
10. Explain the basic mechanism of respiration (Ja-17, May-17)
11. Mention the function of respiratory system.
12. Write down peculiarities of pulmonary circulation **(CMU**:Nov-24, 21, 19) (May-18)
13. State the dynamics of fluid exchange through the lung capillaries. **(CMU:**Nov-20) (May-18)
14. Show in a diagram the different pressure changes during respiration (Nov-18)
15. Show in a diagram about volume & pressure changes that occur during quite breathing. **(CMU:**May-20)
16. What do you mean by the physiological shunt and physiological dead space. **(CMU:**Nov-24, 21)
    What is pulmonary & alveolar ventilation **(CMU:** Nov-22)

Respiratory Membrane

1. Draw and level the respiratory membrane. **(CMU: May-25**, 23) **(Nov-25)** (May-19, Ja-05,06,11,13, Ju-07,14, Nov-15, Ja-16) How gas exchange occur through this membrane. **(CMU:**Nov-22, 19, May-19) (Ja-15, Nov-15, Ja-16)
2. List the factors influencing the gaseous diffusion through the respiratory membrane **(CMU:May-25,** 23) **(Nov-25)** (Ja-07, 08).
3. What is dead space? Write down the differences between anatomical and physiological dead space. Mention their importance's. (Ja-07,10, Ju-09)
4. State the composition & partial pressure of respiratory gases. (Ja-15)
5. Name the layers of alveolo capillary membrance and factors affecting gaseous exchange across it. **(CMU:** Nov-24)
6. **S/N:** Respiratory Membrane **(CMU:**Nov-20)

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What is difficing capacity of lung? **(CMU: Nov-25)**

Surfactant

1. Write down the composition and function of surfactant (May-15, 16). What do you mean by respiratory distress syndrome of new born? (Ja-08)
2. State the importance of surfactant in prevention of IRDS. (Ju-10, 11, May-19)
3. List the factors causing lung collapse. How collapsing tendency can be prevented?
4. **S/N:** Surfactant **(CMU:** Nov-21, 20, Nov-19, May-22, 19) (May-18)

Lung volume & capacities

1. Name the lung volume and capacities with their normal values. **(CMU:**Nov-24, May-20) (Ja-05, 08, 11, Jan-16)
2. Draw a Spirogram to show the different lung volume and capacities(Ju-06,09, 13, Ju-15, May-18, Ja-06, 11,13 Ju-06, 09)
3. How do you measure residual volume? (Ja-12,13) Mention their physiological importance. **(CMU:**May-19) (Ju-08, 07, Ja-10, 14, Nov-15, Ja-16)
4. Explain why FEV1 is decreased in obstructive lung disease but not restrictive lung disease? (Ja-05,10)
5. Define vital capacity **(CMU:** Nov-22) (Nov-17)Name the factors influencing vital capacity. (May-17, Nov-17, Ju-06,14) with physiological importance.(Nov-16, May-19)
6. State in a table the different constituts of inspired alveolar air. (Ja-10,13, Ju-08)
7. Define diffusion capacity. What are the factors influencing diffusion capacity of the lung. (Ja-05)
8. Mention the importance of residual volume & vital capacity. **(CMU:**May-20)
9. Name the lung function tests of clinical practice. State its clinical importance with interpretation. **(CMU: May-25)**
10. List the functions of lungs. **(CMU:**May-19) (Ju-14, May-15, Ja-17)
11. Write importance of tidal volume? (Ja-15) Mention the pulmonary function tests.**(CMU:**May-24)
12. What is FEV1 and peak expiratory flow rate ? (May-16)
13. State the importance of FEV1 **(CMU:** Nov-24) (Ja-06, 11, 13, Ju-06, 09, 07, May-17)
14. Draw and level spirogram. **(CMU:**Nov-22, 20) **(CMU: Nov-25)**
15. What do you mean by the dead space Air volume? How do you measure this volume by using Bohr's method. **(CMU:** May-21)
16. How obstructive airway diseases can be differentiated from restrictive airway disease?**(CMU:May-25,24)**
17. **S/N:** FEV_1 (Nov-17)
18. Define FVC and FEV. **(CMU: May-25)**

Transport of O_2 & CO_2

1. How O_2 and CO_2 are transported in the blood? **(CMU:**Nov-22) (Ja-06,13,14)
2. How O_2 is transported (Ja-17) from lungs to tissue. **(CMU:**Nov-20) (Ja-13, Ju-14, May-15).
3. Discuss the gaseous exchange in lung & tissue (Ju-15, Nov-16)
4. Mechanism of gaseous exchange in Alveolus. **(CMU:**May-21)
5. State the composition & partial pressure of respiratory gases that enter & leave the lungs. (Ju-10, May-15)
6. What is Bohr Effect? **(CMU:**Nov-20) What is it's physiological significance? (Ja-05, 10, Ju-06, 11)
7. Draw and level Oxyhaemoglobin dissociation curve. **(CMU:**May-22, 21, 19) (Ju-07, 08,12 Ja-09, 12, Nov-15,16,17,18, Ja-16, 17, May-16, May-17, 19) **(CMU:**May-19) why this curve is sigmoid shape? (Ju-07, May-15, Ju-15, Nov-15,18, Ja-16) Name the parts with importance **(CMU:**May-19) (Nov-16,18, May-19)
8. Name the factors that influence the shifting of the curve and mention the significance of shifting. (Ja-05,09,12,15, Ju-05, May-15,17, Ju-15, Ja-17) Write down its importance (Nov-17)
9. Write down the fate of CO_2 in blood. (Ja-11)

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10. Name the different forms in which O_2 is carried to blood. (Ja-05)
11. Describe the manner in which O_2 flows downhill from lung to tissue and CO_2 from tissue to lung(Ju-10). Name the factors that influence these transports. (Ja-07, Ju-08)
12. Administration of O_2 in a patient with severe respiratory failure may cause death - why? (Ja-06, 07)
13. Name the forms in which CO_2 transported in blood. Show in a diagram the mechanism and physiological significance of chloride shift. (Ju-09, Ja-14) **(CMU: May-25)**
14. What do you mean by P^{50}? What are the significances of increased and decreased P^{50}? **(CMU:**May-21, 22) (Ja-08, May-16)
15. **Short note:**
    a) Haldane effect. **(CMU:**May-19) (Ju-09, Ju-15, Ja-16, Nov-16, May-16, 19)
    b) What is physiological shunt (Ju-13, Ju-15, May-18)
    c) O_2-Hb curve (Ju-14) d) Dead space air. E) Chloride shift mechanism. **(CMU: Nov-25)**
16. Define hearing Brever inflation reflex and explain. How does it control rate and depth of respiration. **(CMU: Nov-25)**

Respiratory Center

1. Name the higher respiratory center **(CMU:**May-19) (Nov-18) with their location and function **(CMU: May-25,** Nov-19) (Ja-06, 08, 14, Ju-09, 13, May-15)
2. Mention organization of respiratory center. **(CMU:** May-24)
3. Name the chemical and non-chemical that regulate respiration. **(CMU:**May-23) (Nov-18)
4. Mention the stimuli that affect the respiratory center **(CMU:**May-20, 22) (Ja-12, Ju-11) CO_2 has direct effect in stimulating the chemo sensitive area - explain?
5. Describe the role of pre bottzingers complex in producing spontaneous breathing. (Ja-06)
6. Give a brief account on the neural & chemical regulation of respiration **(CMU:**Nov-19) (Ju-05, 07, 13, May-15) Chemical regulation of respiration. **(CMU:**May-24, 20)
7. Name the receptors that are present in the lungs. Explain and illustrate the changes that occur during and before exercise. (Ju-06)
8. What is rhythmic or periodic breathing. Describe the mechanism of rhythmic breathing. **(CMU:** Nov-20, May-22) (Ju-06,09, Ja-14, 15, May-15, Nov-17)
   Which is maintained by respiratory center **(CMU:**Nov-23)
9. What do you mean by respiratory acidosis and alkalosis? What are the effects of hypercapnia in the body. (Ja-06)
10. Briefly discuss the effects of exercise upon respiration (Ju-06, Nov-16)
11. Describe the rhythmical regulation of respiration. (Ju-14, May-16, 18)/ RAMP Signal (May-16)
12. Write in short about the reflex that prevent lung from hyperinflation **(CMU:**Nov-23)
13. **SN:-** a) Apnea **(CMU:**May-21) (Ja-09, 10, Ju-09)
    b) Asphyxia (Ja-06)
    c) Periodic breathing. (Ju-06,12, Nov-16)
    d) Hypercapnia (Ja-06)
    e) Chyene stroke respiration
14. What is Ramp signal **(CMU:May-25,** 21) (May-18) with its comportance. **(CMU: Nov-25)**
15. What is Hering reflex (May-18)
16. Briefly discuss about Bohr effect (May-18)
17. What are the changes that occur in respiratory system during exercise. **(CMU:**May-20)
18. CO_2 has direct effect in stimulating the chemo-sensitive area explain. **(CMU:**Nov-21)
19. Draw and label respiratory center. **(CMU: May-25)**

Hypoxia

Define hypoxia and **(CMU:**May-20) classify it with example **(CMU:**May-21, Nov-22) (Ju-06,10, May-16).

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2. Administration of O_2 may stop respiration and even causes death in patient with sev respiratory failure. Why? (Ja-05, 06, 08, 09, 10, Ju-06) Why there is hypoxia in anemia? (Ja-0
3. **SN:** a) Hypoxia. (Ja-11, 12, 14, Ju-13, Nov-17) **(CMU: Nov-25)**
   b) Cyanosis. (Ja-11,13,14,15, Ju-08,09,14, Nov-15, 16, 17, May-19)
   c) Hering Breuer reflexes? **(CMU:**May-20) (Ja-06,Ju-12) d) Arterial blood gas (Ja-09,1

GASTROINTESTINAL SYSTEM

Hormone

1. Name the local hormones (May-18) (Ju-15) secreted from the GI tract. **(CMU:**Nov-20, May- State the function of them. (Ju-05, Ja-10, 06, 08, 12, May-15, Ja-17)
2. Which hormone influences GIT movement (Ju-12, Nov-18)
3. Functions of Gastrin (Ja-17, May-18, 19) **(CMU:**May-19) and cholecystokinin **(CMU:**Nov- (May-18) Secretin **(CMU:**Nov-21, May-19)
4. **SN:** Gastrin. **(CMU:** May-20)

Digestive juice

1. Name the different digestive juices with their pH & daily secretion. (Ja-05)
2. Name the principal digestive enzymes (Ju-12). Give their precursors, substrates and products action of enzyme. (Ja-07, 08, 12, Ju-11)
3. Give the composition (Ja-14) and function of saliva. **(CMU**: May-23, 21) (Ja-15, Nov-15,16 16) Name the lipolytic enzymes.(Ja-13)
4. List the main constituents of gastric juice. **(CMU:**May-23, 20) (Ju-11, Ja-14) Show in a diag the mechanism of HCl secretion. **(CMU**:May-22, 21, Nov-22, 19) (Ja-05, 14 Ju-07, 09, 14)
5. Name the cells of stomach with their secretion & function (Ja-13)
6. Describe the phase of gastric juice secretion(Ju-12). How gastric juice secretion is regulated? (Ja-11) & maintain its cells secretion. (Ju-11) Mechanism of HCl secretion. (Nov-17, 18)
7. Write the composition of bile. Role of bile in fat digestion digestive system? **(CMU:** Nov- (Ja-05, Ju-10, 11,12, May-15, Ju-15) What is enterohepatic circulation? (May-16)
8. Name the bile salts with their functions. How the secretion of bile is controlled. (Ju-07,Ja-06,1
9. Name the pancreatic digestive enzymes **(CMU:May-25,** 19). (Ja-16, Nov-16, May-19), Expl the regulation of pancreatic secretion. **(CMU:** Nov-20, May-20) (Ja-10, Ju-09,14, May-16, 18
10. List the function of liver (Ja-17, M-18). Give the clinically important liver function test w interpretation & importance.(Ju-12, 13, Nov-17)
11. Why there is anaemia in gastric atrophy?
12. Outline the events that occurs during the digestion of CHO in different part of GI tract. (Ja-12)
13. Write down the role of bile salts in fat absorption & digestion. **(CMU:** May-21, 20)
14. How saliva maintain the oral hygiene? (May-19)
15. Name the digestive enzyme of succus entericus. (Ja-15, 17 Nov-15)
16. Write down the composition of pancreatic secretion. **(CMU: May-25,** 21) (Ju-14)
17. What are the functions of bile ? What is ferrying action? (Nov-15)
18. Name different proteolytic enzyme with their actions on substrove in GIT? **(CMU:**May-20) (Ju-15) Mention the enzyme of protein digestion in different part of GIT.
19. Name the enzymes in succus entericus and describe the mechanism of their Action (Ja-17)
20. Define and classify jaundice (Ja-17)
21. Discuss the role of bile in digestion & absorption.
22. Lactose intolerance
23. Function of secretin (May-19)
24. Discuss about regulation of saliva secretion **(CMU:** May-19)
25. List the function of gallbladder **(CMU:**May-21)

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26. What are the carbohydrate splitting enzyme from different part of GIT. Mechanism of carbohydrate absoption throught GIT. **(CMU:** May-22)
27. **SN:** b) Enterohepatic circulation. (Ju-12 Ja-07, 08, Ja-16)
    c) Achlorhydria. (Ja-06, 14)
    d) Alkaline tide. (Ju-12, 13)/ Post prandial alkaline tide **(CMU:** Nov-22) (Nov-18)
    e) Mucosal barrier. (Ju-08)
    f) Cholagogues (Ju-15).
    g) Hepatic bile (Ju-13)
    h) Pancreatic juice (Ja-14)
    i) Choleretic (Ju-15)
    j) Segmentation (Ja-16)
    k) Pyloric pump (Ja-16)
    l) Gastric juice **(CMU:** May-19) (Nov-16, May-17)
    m) Micelle (May-17)
    n) Saliva (Nov-17)
    o) Laclose intolerance. (Nov-18)
    p) Gastric gland **(CMU:** May-21)

Movement

1. State the motor function of stomach **(CMU:**May-20, 22) (May-15). Name the factors which promote and inhibit the emptying of stomach. **(CMU:**Nov-21) (Ja-08, 11)
2. Describe in short the mechanism of swallowing. What is achalasia cardia? (Ja-10) pharyngeal stage of swallowing **(CMU:** May-23) (Nov-22)
3. How emptying of the stomach contents occur and what are the factors that cause emptying? (Ju-06)
4. Write the function of gall bladder. What are the factors that influence gall bladder emptying? (Ja-06)
5. Name the different movement **(CMU:** Nov-22, 21,19) (May-17) of GI tract **(CMU:** Nov-24, 17). State in brief any two of them. **(CMU:**May-23) (Ja-07,09, Ju-10,13,14)
6. Name the gastrointestinal reflexes (May-17, Nov-18). Show in diagram the control of defecations reflex. (Ja-12, 15, 17, Ju-10,14, May-16)
7. List the function of large intestines. **(CMU:** Nov-22, 21, 20) (Ju-15, May-17)
   Colon (May-16, 18)
8. Give an account of electrical activity of GIT. How are they affected by sympathetic and parasympathetic stimulus? (May-16) Brief in short about BER. **(CMU:**Nov-23)
9. State the functions of gastrin & secretin.
10. Show in a diagram of neuronal control of gut wall. (Nov-18)
11. List the function of myenteric and sub mucosal plexus (Nov-18)
12. Write shortly/briefly about defaecation reflex **(CMU:** Nov-24, 22, 20) (May-18)
13. Briefly discuss about segmentation contraction (Nov-18)
14. **SN:** a) Gastrin (Ju-14, Nov-17)
    b) Micelle (Ja-15, Nov-16, 18)
    c) Peristalsis **(CMU:** Nov-23, 20) (Ju-14, Nov-16, May-18)
    d) GIT reflex **(CMU:**May-23, Ja, Nov-15)
    e) Defaecation reflex **(CMU:**Nov-23)(Nov-16, 17, May-19) **(CMU:** May-21, 19)
    f) Haustration (May, Nov-18, 22)
    g) Segmentation (May-19)
    h) Achalasia **(CMU:** May-22)

Digestion & Absorption

1. Def. Digestion (Ju-13), Digestion and absorption of CHO (Ju-09, Ja-14) Protein, fat (Nov-16) (Ja-10,13, Ju-13)
2. Explain protein digestion in intestine, Name the enzymes that digest protein. (Ja-11)
3. Summarize digestion and absorption of starch in the alimentary tract **(CMU:**Nov-19)
4. What is alkalitide **(CMU:** Nov-22)

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